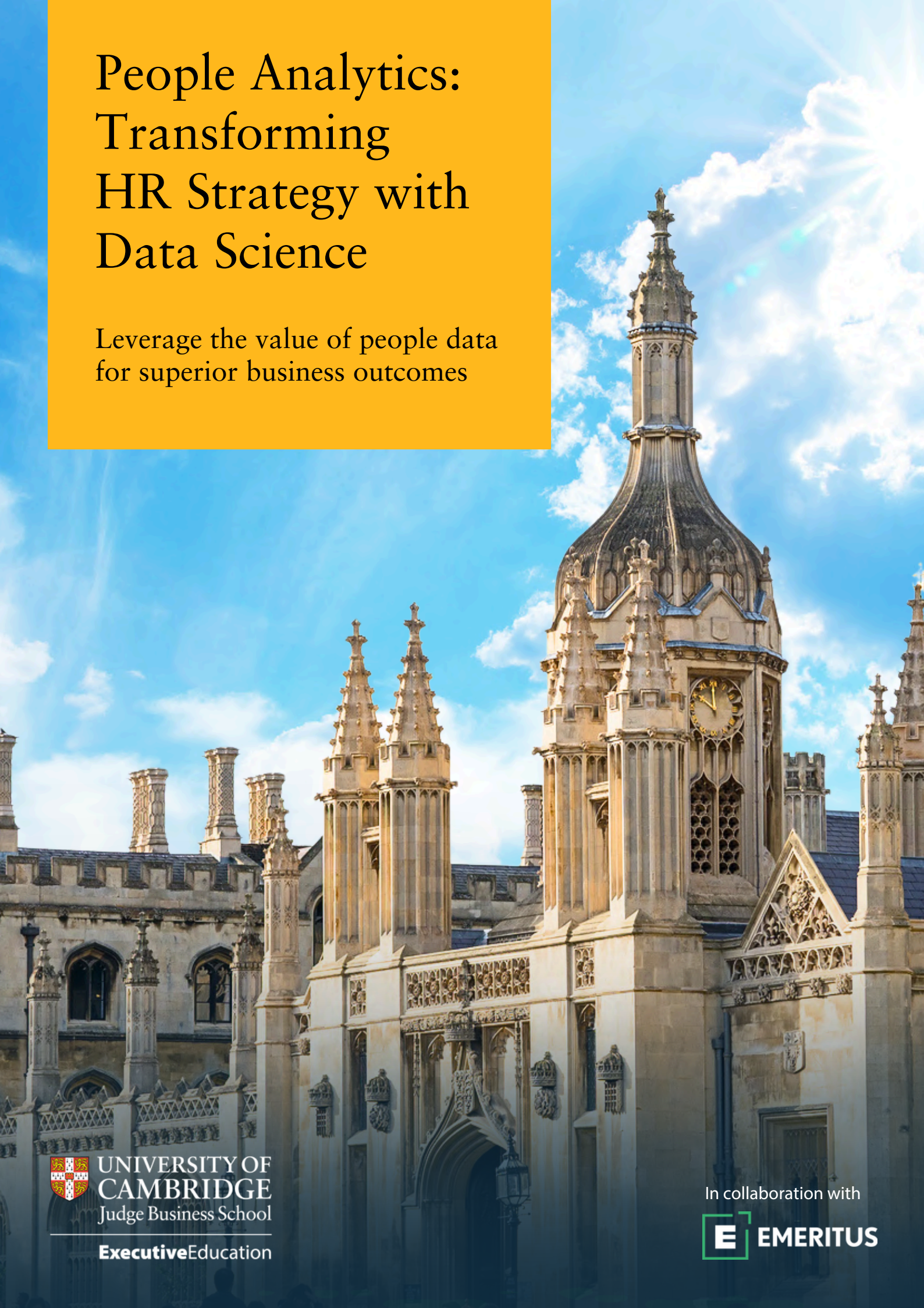
People Analytics: Transforming HR Strategy with Data Science

Leverage the value of people data for superior business outcomes



UNIVERSITY OF CAMBRIDGE
Judge Business School

ExecutiveEducation

In collaboration with

EMERITUS

Overview

Today's leaders operate in an increasingly dynamic and data-rich environment. The effective combination of strategy and science has therefore become indispensable to delivering better outcomes for both businesses and their people. However, many business leaders and HR professionals are relatively new to data-driven techniques and are often limited to descriptive reporting rather than leveraging data to inform strategy and predict future outcomes.

The **People Analytics: Transforming HR Strategy with Data Science programme** from Cambridge Judge Business School Executive Education is an intensive eight-week learning experience for leaders and professionals who want to build or scale their people analytics functions and drive data-informed transformation within their organisations. Integrating the latest findings from in-depth research by the University of Cambridge Psychometrics Centre and practical insights from industry experts, this programme enables you to define and position a people analytics function, translate organisational strategy into actionable initiatives and apply scientific methods to solve workforce challenges.

Designed for senior HR leaders, this programme moves beyond dashboards and reporting to help you build a strategic, decision-enabling people analytics function. You will learn how to connect people insights directly to business performance, organisational design and leadership decisions, enabling you to influence the C-suite with evidence-based recommendations.

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HR leaders believe that if they don't take action to improve HR's approach to technology, their function's effectiveness will decrease.

Source: Gartner

Price
£2,100

Singapore residents who wish to enrol in this programme will be charged GST.

Duration
8 Weeks, Online
6 hours per week



Learning outcomes

By the end of this programme, you will be able to:

- Establish the setup and define the mission of a people analytics function
- Translate organisational strategy into effective people analytics initiatives using stakeholder insight and data
- Identify the root causes of people's challenges to align problems with appropriate solutions
- Design people analytics experiments to test hypotheses and generate actionable insights
- Apply psychometric tools to enhance people analytics and support decision-making
- Leverage data to analyse past trends, interpret current conditions and predict future outcomes
- Recommend evidence-based organisational actions informed by predictive modelling
- Promote the responsible and ethical use of people analytics within organisational and societal contexts

People Analytics with Cambridge Judge Business School Executive Education



The **People Analytics: Transforming HR Strategy with Data Science programme,** delivered in collaboration with the University of Cambridge's The Psychometrics Centre, provides a practical and research-driven approach to building people analytics capability.

Guided by expert faculty from Cambridge Judge Business School and industry practitioners, you will learn to define and position a people analytics function, translate strategy into data-driven initiatives and generate actionable insights through experimentation, psychometrics and predictive modelling, as part of a global cohort of professionals.

Ranked among the world's leading business schools, Cambridge Judge Business School Executive Education combines academic excellence with real-world relevance to equip leaders with the skills, insights and confidence needed to drive innovation and make evidence-based decisions.



"In this online learning experience, you will learn the fundamentals of people analytics and transform how your business works with its most precious resource — its people."

VESSELIN POPOV

Executive Director, The Psychometrics Centre, Cambridge Judge Business School

Programme highlights

The programme includes a mix of live and recorded sessions with faculty and guest speakers, applied exercises, interactive discussions and psychometric instruments. You are expected to dedicate six hours per week over the eight-week programme duration.

Here are some key takeaways you can expect from the People Analytics programme.



Setting up a people analytics function

- Clarify what it takes in practice to set up a people analytics function and how to best position it within your organisation.
- Refine the ways you collect, analyse and maintain organisational databases.
- Equip your function with the necessary techniques to tackle the most salient challenges.



Generating actionable insights

- Apply scientific research methodologies to enhance information quality.
- Experiment with confidence in complex and dynamic organisational settings.



Moving from description to prediction

- Explore how prediction models are designed and how they operate to enable more accurate forecasting.
- Be inspired to shape the future through accountable and data-driven leadership.



Using data to solve problems

- Learn the language of data science to better manage analysts working with people data.
- Integrate psychological theory with strategic imperatives to significantly improve project outcomes.



Action planning for your organisation

- Reflect, discuss and consolidate each week's learning.
- Get practical guidance and support in implementing people analytics.



Who is this programme for?

This programme is designed for professionals looking to leverage people analytics to drive measurable organisational impact, shape a data-driven workforce strategy and chart a clear vision for the future. It supports both senior HR leaders seeking strategic maturity and emerging professionals building capability.

This programme is ideal for:

- **Senior HR leaders and strategic decision-makers,**
  including CHROs, CPOs, HR Directors and HRBPs, who aim to strengthen their people analytics strategy and drive organisation-wide, evidence-based decision-making.
- **People analytics and HR professionals**
  seeking to deepen their strategic understanding of people analytics and apply advanced insights to practical, business-critical workforce challenges.
- **Consultants, transformation specialists, advisors and cross-functional project leaders**
  who require analytical fluency and strategic framing to deliver high-impact, data-driven solutions across organisational initiatives.
- **Employee representatives and business leaders adopting data-driven work practices**
  who want to develop a sophisticated understanding of how big data and AI are reshaping workforce dynamics, decision-making and organisational design.

You do not need a technical background. The programme equips you to lead analysts, interpret complex models and translate insights into strategic action.

The learning objectives of this programme are applicable across functions, industries and geographies. From traditionally data-rich environments, including finance, telecoms and technology, to emerging applications of data science in government and the public sector, people analytics offers significant potential. You will gain the knowledge and perspective needed to translate these opportunities into tangible organisational outcomes.

Learning journey

Module 1: Strategic Positioning

Understand how to assess and position a people analytics function within your organisation, and evaluate its maturity, impact and alignment with strategic priorities.

- Benchmark your organisation's people analytics capability against relevant industry examples.
- Assess professional maturity and functional maturity in relation to people analytics capabilities.
- Identify opportunities for a people analytics function to inform, optimise and align strategy with corporate objectives.
- Evaluate the current and future state of people analytics capability within your organisation.

Module 2: Leveraging Internal Resources

Explore how to identify, structure and scope people analytics initiatives by effectively using internal data, infrastructure and stakeholder insight.

- Analyse stakeholder roles, goals and evidence needs in relation to people analytics initiatives.
- Evaluate people analytics projects in relation to project type, organisational maturity and strategic impact.
- Apply top-down and bottom-up approaches to identify people analytics opportunities.
- Diagram your current data warehousing operations.
- Outline a people analytics project by analysing stakeholders, data, infrastructure and organisational context.

Module 3: Root Cause Analysis and Project Tactics

Develop a structured approach to diagnosing workforce challenges and prioritising projects based on underlying organisational issues.

- Sequence people analytics projects under capacity constraints.
- Formulate counterintuitive hypotheses to investigate workforce challenges.
- Determine organisational symptoms, first-level problems, higher-level problems and root causes.
- Prioritise people analytics projects based on root cause analyses.

Module 4: Experiment Design for People Analytics

Learn how to design rigorous experiments that test hypotheses and generate reliable, actionable insights for organisational decision-making.

- Draft hypotheses and determine the variables for a people analytics experiment.
- Choose an appropriate experiment design to test a hypothesis.
- Identify ways that statistical analysis helps to draw insights from data.
- Describe how scientific methods can resolve practical challenges.
- Design a defensible experiment to address a strategic people challenge.

Learning journey

Module 5: Psychometric Testing

Examine how psychometric tools can enhance people analytics by providing deeper insights into individual and organisational behaviour.

- Explore why psychometrics matters and where it adds value in an organisation.
- Evaluate the quality of psychometric tests.
- Differentiate between ability and personality assessments used in organisational contexts.
- Detect response bias risks in the design and administration of psychometric assessment.
- Recommend suitable psychometric tests to address organisational challenges.

Module 6: Organisational Data Science (Part 1)

Build foundational capabilities in interpreting organisational data and translating analytical outputs into strategic insights.

- Analyse how organisational mental models shape workforce data.
- Evaluate how structured interviewing influences the strategic direction of analytics initiatives.
- Interpret organisational text as a source of behavioural and strategic signals.
- Translate data science concepts, model outputs and analytical approaches in a strategic context.
- Frame a structured plan for analytical work that supports an organisational initiative.

Module 7: Organisational Data Science (Part 2)

Understand how predictive modelling and AI are applied in people analytics and how to communicate insights effectively to stakeholders.

- Distinguish between machine learning and AI problem types.
- Explain classical use cases and the workflow of predictive modelling.
- Communicate predictive insights to stakeholders.

Module 8: Transforming the Workplace

Explore the ethical, organisational and societal implications of people analytics, and define a responsible approach to its application.

- Assess how societal and technological change shape the ethical application of people analytics.
- Recommend governance and leadership approaches to mitigate risks associated with predictive analytics and AI.
- Articulate a principled leadership stance on the responsible use of people analytics technologies.

Practical tools and frameworks



Build the capability to assess and elevate your organisation's people analytics maturity using proven frameworks, practical tools and real-world models, enabling more informed decisions and sustained strategic impact.

Gartner's IT Maturity Model
A structured assessment tool, this model offers benchmarks and actionable road maps for evolving IT capabilities across domains, including CIO, data and analytics and I&O.

Talent Analytics Framework (Gartner)
Designed to embed analytics into everyday HR and business workflows, this framework helps HR leaders deliver outcomes from hiring to retention through scalable, decision-driven insights.

Gartner Corporate Performance Management (CPM) Framework
This framework is a holistic system that aligns organisational strategy, metrics and processes, including planning, forecasting and reporting, with analytical applications to consistently drive enterprise performance.

Orpheus Business Personality Inventory (OBPI)
A tool developed by the Cambridge Psychometrics Centre, the OBPI scientifically assesses workplace preferences and integrity, generating narrative reports for recruitment, coaching and development while providing hands-on insight into psychometric testing and its strategic value in people analytics.

STAR(L) Interview Model and TOTE Scheme
This method provides a structured approach to conducting behavioural interviews and is a proven way to extract actionable information from individuals in the field.

Text analysis: video demonstration
In this recorded walkthrough, the faculty demonstrate text analysis using a Jupyter Notebook to help you understand and interpret the outcomes.

Predictive model analysis: video demonstration
In this recorded walkthrough, the faculty guide you through a predictive modelling exercise step by step.



The learning experience

Practical application
Apply core people analytics concepts to strengthen your organisation's analytics maturity while gaining hands-on experience with tools and frameworks to drive data-informed workforce decisions.

Goal setting
Stay on track with your learning through weekly checkpoints, including video lectures, assignments and quizzes, designed to reinforce your progress.

Faculty live sessions
Engage with Cambridge Judge Business School faculty through live, interactive webinars designed to deepen your understanding of key people analytics concepts and their real-world application.

Practical tools and frameworks
Apply proven models, including Gartner's IT Maturity Model and the OBPI, to assess capability, structure initiatives and enable evidence-based people analytics decision-making.

Case studies and industry examples
Explore real-world case studies, including Google's Project Oxygen and performance forecasting in banking, to understand how organisations solve workforce challenges using people analytics.

Guest speakers
Gain insights from industry experts as they share practical perspectives on applying people analytics, from predictive modelling to workforce strategy, in complex organisational contexts.

Continuous programme access
Continue to have access to programme videos and learning materials for up to one year from the programme start date.

Case studies

Gain critical insights from real-world examples of how organisations across sectors apply people analytics to solve complex workforce challenges and drive strategic outcomes.



Explore how a European airline addressed organisational tensions and enhanced the impact of its people analytics function.



Examine how a software provider applied psychometric data to improve the accuracy of predictive HR models.



Analyse how an Asian bank improved trader performance forecasting while advancing team capabilities in natural language processing, AI and decision-making.



Investigate how a sports clothing brand used data aggregation and stakeholder management to resolve key performance challenges.



Identify how a US bank uncovered flaws in performance ratings and increased HR's strategic value through predictive analytics.



Uncover how a food processing organisation leveraged root cause analysis to better understand attrition and inform business strategy.



Discover how a multinational technology company used a data-driven experiment to reshape perceptions of management and workplace effectiveness.



Assess how a Japanese digital marketing organisation introduced new processes and a predictive model to reduce recruitment bias.

Programme faculty



PETER ROMERO

People Analytics Lead, The Psychometrics People Analytics Lead,
The Psychometrics Centre, Cambridge Judge Business School

Peter Romero brings over 20 years of global experience across research and consulting, having worked with leading organisations, including Gartner and SHL. His background spans research and development, as well as advisory roles supporting complex, large-scale people analytics initiatives.

He advises senior leaders on building and scaling people analytics functions, helping organisations translate complex data into strategic workforce decisions. His work focuses on applying advanced analytics, AI-driven modelling and behavioural insights to drive measurable organisational impact.



LUNING SUN

Research Director, The Psychometrics Centre;
Research Associate, Cambridge Judge Business School

Luning Sun is a leading expert in psychometrics and behavioural measurement, with extensive experience in developing advanced assessment tools and leading international research collaborations. He brings deep expertise in translating complex behavioural data into actionable insights for organisations.

Sun's research focuses on applying cutting-edge psychometric techniques to real-world organisational challenges, enabling more accurate measurement of talent, performance and potential. His work has contributed to large-scale initiatives focused on cognitive and behavioural assessment frameworks.

Guest speakers



JOHN RUST

Founder, The Psychometrics Centre, Cambridge Judge Business School; Associate Fellow, Leverhulme Centre for the Future of Intelligence, University of Cambridge



ANNE PEARCE

Vice President, Performance and Deployment, Shell



DR. MARIA JESUS BELIZON

Assistant Professor of Human Resource Management, Michael Smurfit Graduate Business School, University College Dublin, Ireland



ANDREAS KYPRIANOU

Global Head of People Reporting and Analytics, Upfield



DR IAN ROTHMANN

Chief Technology Officer, PsychAI Talent Group



DR SANDRA MATZ

David W Zalaznick Associate Professor of Business, Columbia Business School



SUCHI PATHAK

Co-Founder and Chief Psychologist, Aptology Inc



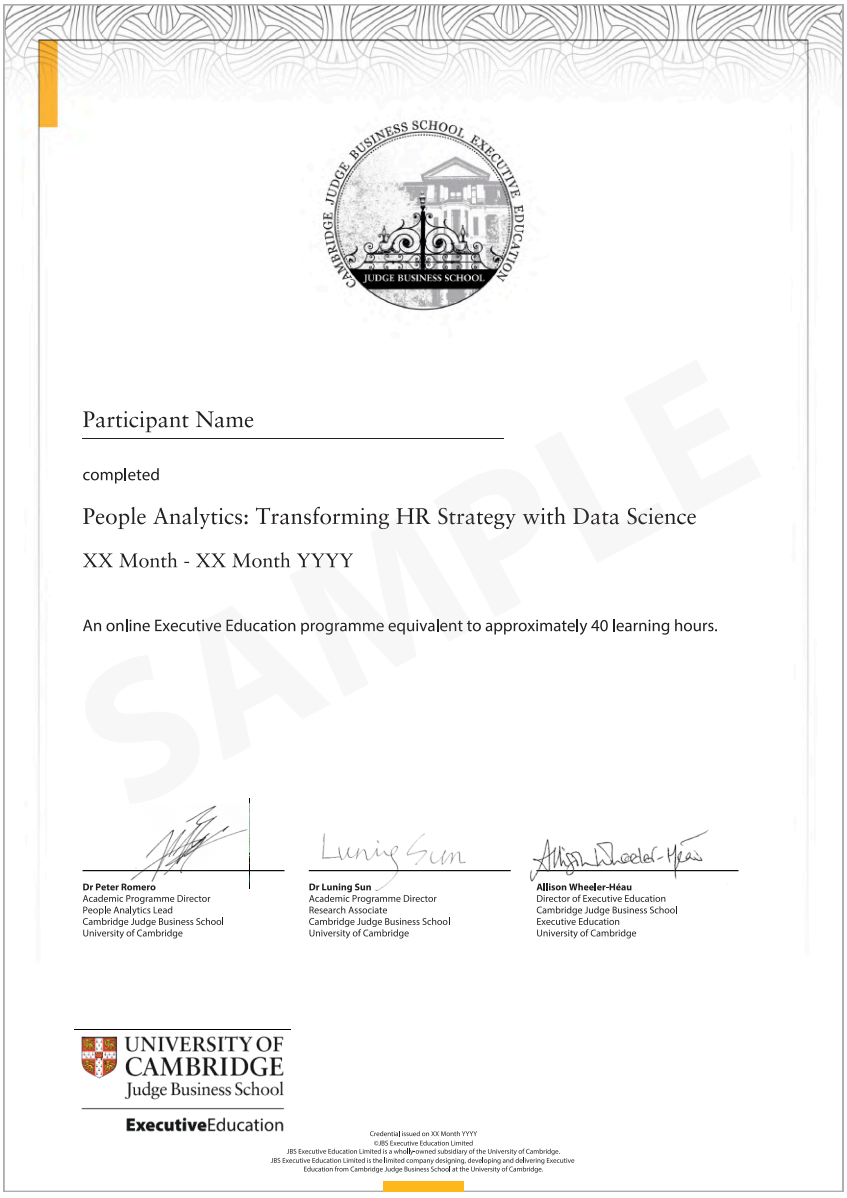
DR ROXANNE LACZO

Senior Consultant, Hogan Corporate Solutions

Note: the faculty list is subject to change.

Certificate

Upon successful completion of the programme, participants will be awarded a digital certificate of completion from Cambridge Judge Business School Executive Education.



About Emeritus

Cambridge Judge Business School Executive Education is collaborating with online education provider Emeritus to offer a portfolio of high-impact online programmes. These programmes leverage Cambridge Judge Business School's thought leadership and technical expertise developed over years of research, teaching and practice.

Helping businesses maximise real-world impact

This programme is being delivered in collaboration with Emeritus.

Prospective participants with questions on the programme are encouraged to contact Emeritus for further information.

Contact us

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